

Descriptive Statistics Primer

Comprehensive Statistical Analysis & Exploratory Data Report of the Titanic Dataset

TOTAL PASSENGERS
891

TOTAL ATTRIBUTES
12 Columns

TOTAL DATA CELLS
10,692

1. Executive Dataset Summary

The Titanic dataset contains information about **891 passengers** across **12 variables** who traveled on the RMS Titanic. It includes demographic details, ticketing information, socio-economic status indicators, and survival outcomes. The dataset consists of both numerical and categorical variables, making it ideal for performing descriptive statistics, missing data evaluation, and distribution modeling.

Column Classifications:

Numeric (7) PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare

Categorical (5) Name, Sex, Ticket, Cabin, Embarked

2. Data Completeness & Missing Value Profile

A key component of data analysis is identifying missing observations across key features. The dataset presents missing values in Age, Cabin, and Embarked.

Column Name	Data Type	Missing Count	Missing Percentage (%)
PassengerId	int64	0	0.00%
Survived	int64	0	0.00%
Pclass	int64	0	0.00%
Name	object	0	0.00%
Sex	object	0	0.00%
Age	float64	177	19.87%
SibSp	int64	0	0.00%
Parch	int64	0	0.00%
Ticket	object	0	0.00%
Fare	float64	0	0.00%
Cabin	object	687	77.10%

Column Name	Data Type	Missing Count	Missing Percentage (%)
Embarked	object	2	0.22%

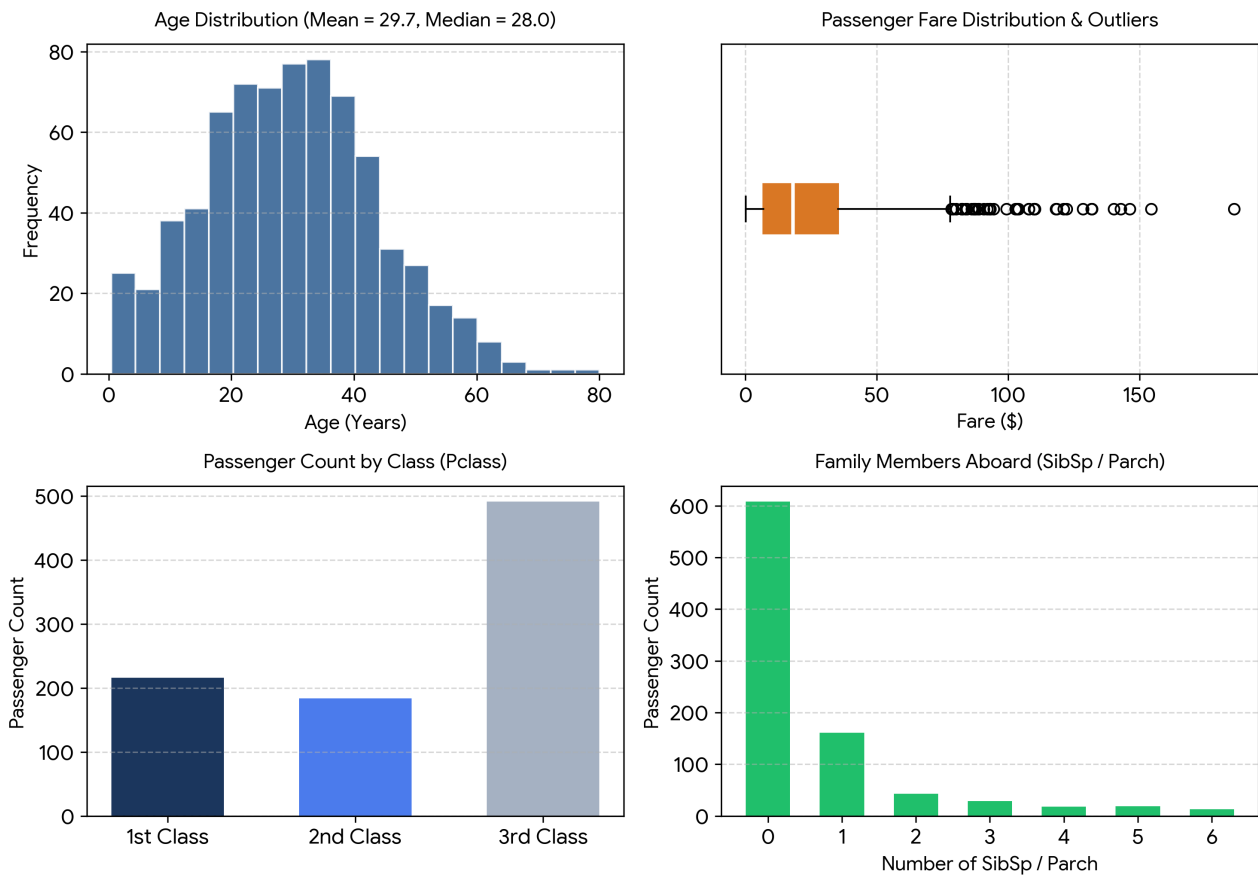
3. Comprehensive Descriptive Statistics

The table below provides central tendency metrics (Mean, Median, Mode), measures of dispersion (Standard Deviation, Range, IQR), and percentile distributions for key numerical features: Age, Fare, SibSp, Parch, and Pclass.

Metric	Age	Fare	SibSp	Parch	Pclass
Count	714	891	891	891	891
Mean	29.699	32.204	0.523	0.382	2.309
Median	28.000	14.454	0.000	0.000	3.000
Mode	24.000	8.050	0.000	0.000	3.000
Std Deviation	14.526	49.693	1.103	0.806	0.836
Minimum	0.420	0.000	0.000	0.000	1.000
25th Percentile (Q1)	20.125	7.910	0.000	0.000	2.000
50th Percentile (Q2)	28.000	14.454	0.000	0.000	3.000
75th Percentile (Q3)	38.000	31.000	1.000	0.000	3.000
Maximum	80.000	512.329	8.000	6.000	3.000
Interquartile Range (IQR)	17.875	23.090	1.000	0.000	1.000

4. Visual Distribution & Trend Analysis

Graphic visualization of key statistical features allows for immediate insight into skewness, outliers, and class distributions across passengers on board.



5. Key Analytical Insights

- **Age Distribution:** Passenger ages range from 0.42 years (~5 months) to 80 years. The mean age of 29.70 years closely matches the median of 28.00 years, indicating a relatively normal distribution with a slight right-skew due to elderly passengers.
- **Fare Skewness & Outliers:** Passenger fares demonstrate extreme right-skewness. While 50% of passengers paid \$14.45 or less (Median), the Mean fare is \$32.20, driven up by high-value outliers up to \$512.33.
- **Passenger Class Distribution:** The majority of passengers traveled in 3rd Class (Pclass = 3, mode = 3), making up over 55% of total passenger density.
- **Family Connections:** Most passengers traveled without siblings/spouses (SibSp median = 0) or parents/children (Parch median = 0). Maximum family groupings aboard reached up to 8 siblings/spouses and 6 parents/children.